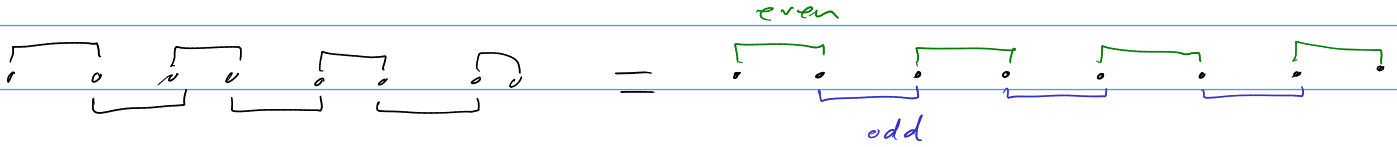


$$h = -it \sum_i \gamma_i \gamma_{i+1} + g \sum_i \gamma_i \gamma_{i+2} \gamma_{i+3} \gamma_{i+4} + \gamma_i \gamma_{i+1} \gamma_{i+2} \gamma_{i+3}$$

split hamiltonian into h_{even} , h_{odd} and rewrite majorana operators into even and odd:

$$a_j := \gamma_{2j}$$

$$b_j := \gamma_{2j+1}$$



$$h = h_{\text{even}} + h_{\text{odd}}$$

$$= -it \left(\sum_j a_j b_j + \sum_j b_j a_{j+1} \right)$$

→ Use Jordan Wigner transformations to map fermionic (majorana) operators to spin operators (Pauli matrices)

$$-it \left(\sum_j i \sigma_j^z + K_j \sigma_j^y K_{j+1} \sigma_{j+1}^x \right)$$

$$= -it \left(\sum_j (i \sigma_j^z + K_j \sigma_j^y K_j \sigma_j^z \sigma_{j+1}^x) \right)$$

$$-it \left(\sum_j i \sigma_j^z + \cancel{K_j} \sigma_j^y \sigma_j^z \sigma_{j+1}^x \right)$$

$= i \sigma_j^x$

$$= -it \left(\sum_j i \sigma_j^z + i \sigma_j^x \sigma_{j+1}^x \right)$$

$$= \sum_j -\cancel{t} \sigma_j^z - \cancel{t} \sigma_j^x \sigma_{j+1}^x = \sum_j -J \sigma_j^x \sigma_{j+1}^x - h \sigma_j^z \quad \text{as expected.}$$

$$1 \times t \quad t = J = h$$

$$a_j = K_j \sigma_j^x$$

$$b_j = K_j \sigma_j^y$$

$$a_j b_j = i \sigma_j^z$$

$$K_j = \prod_{l=1}^{j-1} \sigma_l^z$$

$$\sigma_i \sigma_j = i \epsilon_{ijk} \sigma_k$$

$$g \sum_i r_i r_{i+2} r_{i+3} r_{i+n} = g \sum_j \underbrace{a_j a_{j+1} b_{j+1} a_{j+2}} + \underbrace{b_j b_{j+1} a_{j+2} b_{j+2}}$$

$$\begin{array}{|c|c|c|c|c|} \hline \bullet & \bullet & \bullet & \bullet & \bullet \\ \hline \end{array} = \begin{array}{|c|c|c|c|c|} \hline a_j & b_j & a_{j+1} & a_{j+1} & a_{j+2} & b_{j+2} \\ \hline \end{array}$$

$$a_j a_{j+1} b_{j+1} a_{j+2} =$$

$$K_j \sigma_j^x a_j \sigma_j^z - K_{j+2} \sigma_{j+2}^x = \cancel{K_j \sigma_j^x} a_j \sigma_{j+1}^z \cdot \cancel{K_{j+2} \sigma_{j+2}^x} \cdot \cancel{\sigma_j^z} \sigma_{j+1}^z \sigma_{j+2}^x$$

$$= \sigma_j^x a_j \sigma_{j+1}^z \sigma_j^z \sigma_{j+1}^z \sigma_{j+2}^x$$

$$a_j \sigma_j^x \sigma_j^z \sigma_{j+1}^z \sigma_{j+1}^z \sigma_{j+2}^x$$

$$i(-i \sigma_j^y) \sigma_{j+2}^x = \sigma_j^y \sigma_{j+2}^x$$

$$b_j b_{j+1} a_{j+2} b_{j+2} = \cancel{K_j \sigma_j^y} \cdot \cancel{K_{j+2} \sigma_{j+2}^z} \sigma_{j+1}^y \cdot i \sigma_{j+2}^z$$

$$\sigma_j^y \sigma_j^z \sigma_{j+1}^y i \sigma_{j+2}^z$$

$$\cancel{i \sigma_j^x} \sigma_{j+1}^y \sigma_{j+2}^z = - \sigma_j^x \sigma_{j+1}^y \sigma_{j+2}^z$$

$$g \sum r_i r_{i+2} r_{i+3} r_{i+n} = g \sum \sigma_j^y \sigma_{j+2}^x - \sigma_j^x \sigma_{j+1}^y \sigma_{j+2}^z$$

$$+ \gamma_i \gamma_{i+1} \gamma_{i+2} \gamma_{i+3} = \sum_j \underbrace{a_j b_j a_{j+1} a_{j+2}} + \underbrace{b_j a_{j+1} b_{j+1} b_{j+2}}$$

$$\begin{array}{|c|c|c|c|c|} \hline \cdot & \cdot & \cdot & \cdot & \cdot \\ \hline \end{array} = \begin{array}{|c|c|c|c|c|} \hline a_j & b_j & a_{j+1} & b_{j+1} & a_{j+2} b_{j+2} \\ \hline \end{array}$$

$$\begin{aligned} \underbrace{a_j b_j}_{++} \cdot a_{j+1} a_{j+2} &= i \sigma_j^z \cdot \cancel{\chi_j \sigma_j^z} \sigma_{j+1}^x \cdot \cancel{\chi_j \sigma_j^z} \sigma_{j+1}^z \sigma_{j+2}^x \\ &= i \sigma_j^z \cancel{\sigma_j^z \sigma_j^z} \sigma_{j+1}^x \sigma_{j+1}^z \sigma_{j+2}^x \quad \begin{array}{l} \gamma_j^z \\ \gamma_{j+1}^z \end{array} i \sigma_{j+1}^x \\ &= i \sigma_j^z (-i \sigma_{j+1}^y) \sigma_{j+2}^x = \sigma_j^z \sigma_{j+1}^y \sigma_{j+2}^x \end{aligned}$$

$$\begin{aligned} b_j a_{j+1} b_{j+1} b_{j+2} &= \cancel{\chi_j \sigma_j^y} \cdot i \sigma_{j+1}^z \cdot \cancel{\chi_j \sigma_j^z} \sigma_{j+1}^z \sigma_{j+2}^y \\ &= \sigma_j^y \sigma_j^z (i \cancel{\sigma_{j+1}^z \sigma_{j+1}^z}) \sigma_{j+2}^y \\ &= (i \sigma_j^x)(i)(\sigma_{j+2}^y) = -\sigma_j^x \sigma_{j+2}^y \end{aligned}$$

$$g \sum \gamma_i \gamma_{i+1} \gamma_{i+2} \gamma_{i+3} = g \sum \sigma_j^z \sigma_{j+1}^y \sigma_{j+2}^x - \sigma_j^x \sigma_{j+2}^y$$

→ flipped sign of second term and reversed x, y, z index order
↳ makes sense.

$$h = -t \sum_j (\sigma_j^z + \sigma_j^x \sigma_{j+1}^x) + g \left(\sum_j (\sigma_j^y \sigma_{j+2}^x - \sigma_j^x \sigma_{j+1}^y \sigma_{j+2}^z) + (\sigma_j^z \sigma_{j+1}^y \sigma_{j+2}^x - \sigma_j^x \sigma_{j+2}^y) \right)$$

$$h = -t \sum_j (\sigma_j^z + \sigma_j^x \sigma_{j+1}^x) + g \left(\sum_j (\sigma_j^y \sigma_{j+2}^x - \sigma_j^x \sigma_{j+2}^y) - (\sigma_j^x \sigma_{j+1}^y \sigma_{j+2}^z - \sigma_j^z \sigma_{j+1}^y \sigma_{j+2}^x) \right)$$